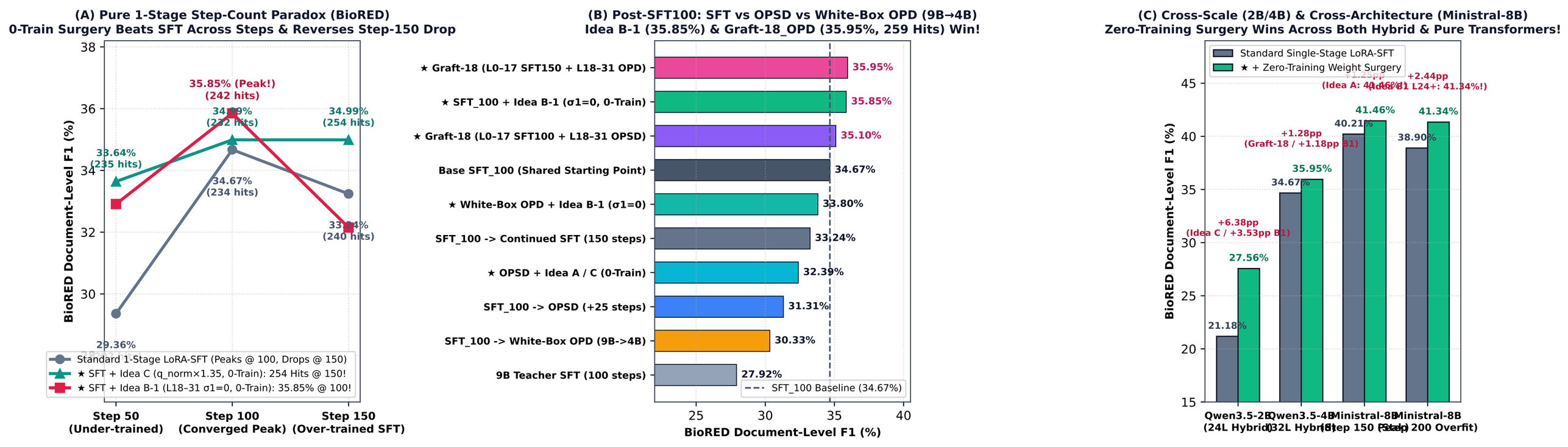
(A) Pure 1-Stage Step-Count Paradox (BioRED)
0-Train Surgery Beats SFT Across Steps & Reverses Step-150 Drop
BioRED Document-Level F1 (%)
38
36
34
32
30
35.85% (Peak!)
(242 hits)
34.99%
(232 hits)
34.99%
(254 hits)
33.64%
(235 hits)
34.67%
(234 hits)
33.24%
(240 hits)
29.36%
Step 50
(Under-trained)
Step 100
(Converged Peak)
Step 150
(Over-trained SFT)
Standard 1-Stage LoRA-SFT (Peaks @ 100, Drops @ 150)
★ SFT + Idea C (q_norm×1.35, 0-Train): 254 Hits @ 150!
★ SFT + Idea B-1 (L18–31 σ1=0, 0-Train): 35.85% @ 100!
(B) Post-SFT100: SFT vs OPSD vs White-Box OPD (9B→4B)
Idea B-1 (35.85%) & Graft-18_OPD (35.95%, 259 Hits) Win!
★ Graft-18 (L0–17 SFT150 + L18–31 OPD)
35.95%
★ SFT_100 + Idea B-1 (σ1=0, 0-Train)
35.85%
★ Graft-18 (L0–17 SFT100 + L18–31 OPSD)
35.10%
Base SFT_100 (Shared Starting Point)
34.67%
★ White-Box OPD + Idea B-1 (σ1=0)
33.80%
SFT_100 -> Continued SFT (150 steps)
33.24%
★ OPSD + Idea A / C (0-Train)
32.39%
SFT_100 -> OPSD (+25 steps)
31.31%
SFT_100 -> White-Box OPD (9B->4B)
30.33%
9B Teacher SFT (100 steps)
27.92%
SFT_100 Baseline (34.67%)
25
30
35
40
BioRED Document-Level F1 (%)
(C) Cross-Scale (2B/4B) & Cross-Architecture (Ministral-8B)
Zero-Training Surgery Wins Across Both Hybrid & Pure Transformers!
Standard Single-Stage LoRA-SFT
★ + Zero-Training Weight Surgery
BioRED Document-Level F1 (%)
45
40
35
30
25
20
15
+6.38pp
(Idea C / +3.53pp B1)
+1.28pp
(Graft-18 / +1.18pp B1)
+2.44pp
(Idea B1 L24+: 41.34%!)
21.18%
27.56%
34.67%
35.95%
40.21%
41.46%
38.90%
41.34%
Qwen3.5-2B
(24L Hybrid)
Qwen3.5-4B
(32L Hybrid)
Ministral-8B
(Step 150 Peak)
Ministral-8B
(Step 200 Overfit)